9 **(a)** Distinguish between *electromotive force e.m.f.* and *potential difference p.d.* using energy considerations.

.....

.....

.....[1]

(b) Fig. 9.1 shows a variable resistor and a fixed resistor R_1 connected to a battery with e.m.f E and internal resistance r. The variable resistor takes a range of values between 0 Ω to R_{max}.

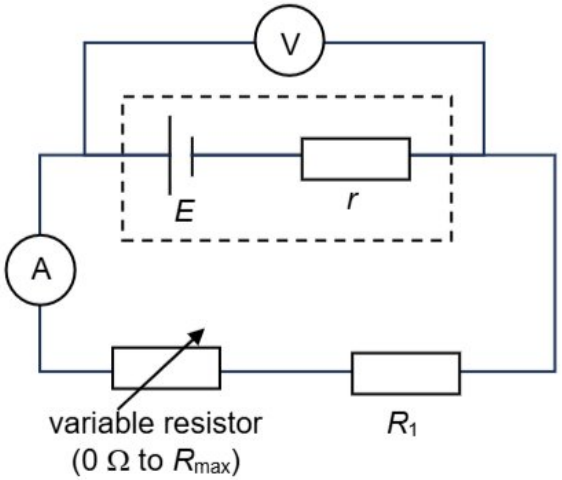


Fig. 9.1

The variation of the voltmeter reading V with current I through the ammeter as the variable resistor is varied over its **full range** of values is shown in Fig. 9.2.

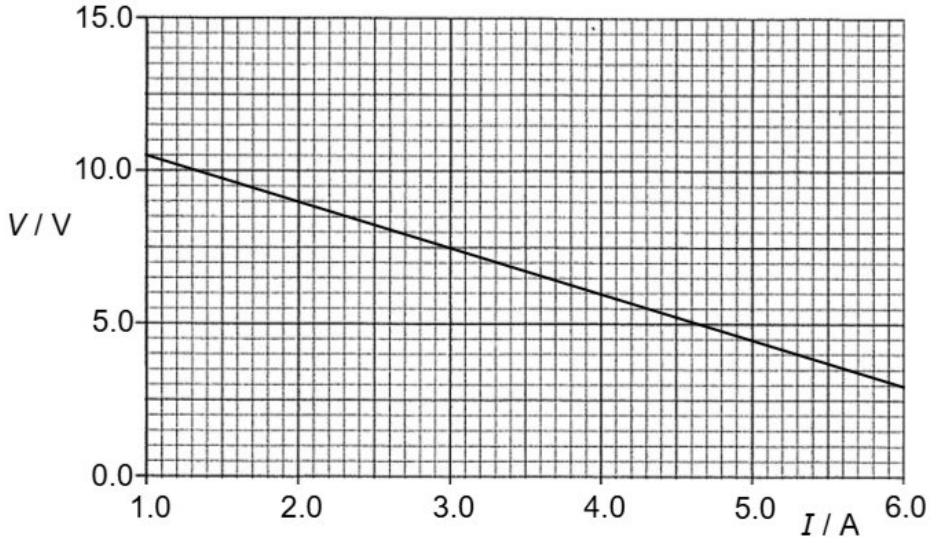


Fig. 9.2

(i) Using the graph, show that that internal resistance r is 1.5 Ω.

[3]

(ii) Calculate the e.m.f. E of the battery.

e.m.f. E = ... V [1]

(iii) Using Fig. 9.2, calculate the maximum resistance of the variable resistor R_{max}. Explain your working.

resistance R_{max} = ... Ω [3]

(c) The variable resistor is now set to 6.0 Ω, and a thermistor is connected in parallel with the variable resistor and R_1. The ammeter is also now connected in series with the thermistor as shown in Fig. 9.3.

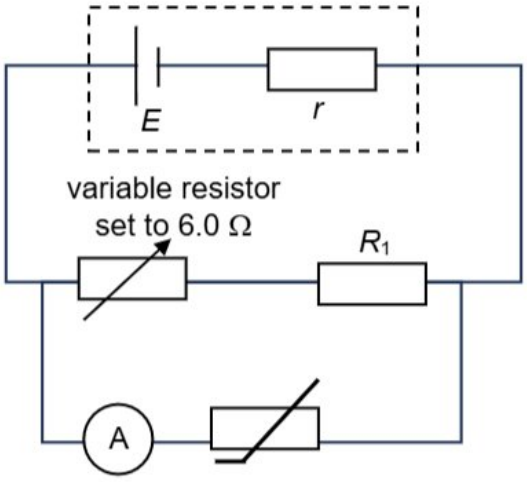


Fig. 9.3

(i) Explain how the potential difference across the thermistor changes when the temperature rises.

.....

.....

.....

.....[2]

(ii) Hence, or otherwise, determine how the current across the thermistor changes when the temperature rises.

.....

.....

.....

.....

.....

.....[3]

(d) Fig. 9.4 shows an alternating input voltage V_{IN} connected to a capacitor C and a resistor R.

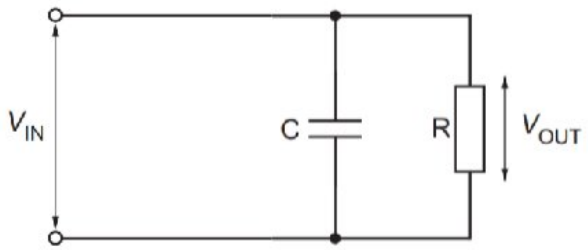


Fig. 9.4

The input voltage V_{IN} is a square wave. Fig 9.5 shows the variation of V_{IN} with time t.

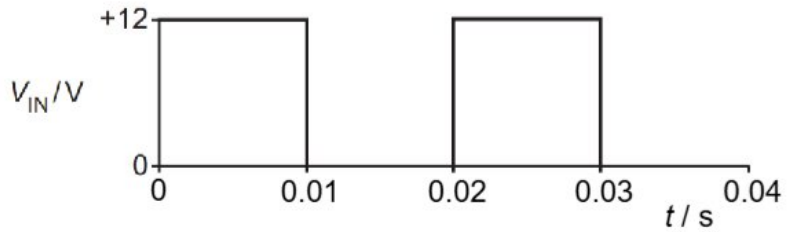


Fig. 9.5

Fig. 9.6 shows the variation of V_{OUT} with time t.

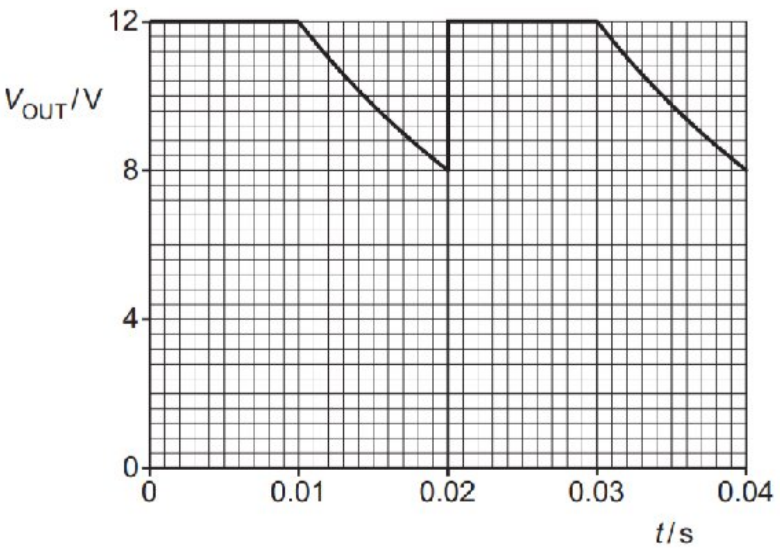


Fig. 9.6

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is 570 μF.

(ii) Determine the resistance of R.

resistance = ... Ω [2]

(iii) Given two capacitors rated 800 μF and a third capacitor of 170 μF, suggest how capacitor *C* of 570 μF can be constructed from these three capacitors. Include a labelled diagram and show your workings clearly.

[3]

[Total: 20]